

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: optical fiber loose tube

Assistant: To better understand your requirements for the optical fiber loose tube:

1. What is the exact use of the polymer in your application? Is it for the loose tube inside the optical fiber cable, or another component (e.g., jacketing, insulation, or cover)?
2. Where will this optical fiber cable be used? For example, is it for outdoor/underground installation, indoor data centers, or another environment?

Once you provide this information, I will ask about wall thickness, preferred base materials, and openness to alternative suggestions.

User: loose tube inside the optical fiber cable, indoor data centers

Assistant: Thank you for clarifying. To proceed:

1. What is the minimum wall thickness of the loose tube inside your optical fiber cable?
2. Do you have any preferred base materials for the loose tube (for example, Polybutylene Terephthalate (PBT), Polyethylene (PE), or others)?
3. Are you open to alternative material suggestions if there is no strict preference for the base material?

These details will help identify the most suitable material options for your application.

User: i dont have any prefrecne thickness of 1.5mm you can take

Assistant: Here are the key material requirements (CTQs) assumed for your indoor optical fiber loose tube application:

1. Minimum wall thickness: 1.5 mm
2. Processing requirement: Extrusion
3. Hydrolysis resistance
4. Low warpage
5. Coefficient of linear thermal expansion (CLTE): $\leq 1.5 \times 10^{-4} \text{ K}^{-1}$
6. Water absorption: 0.005–0.2% (very low)
7. Flexibility
8. Modulus (Tensile or Flexural): $\geq 2200 \text{ MPa}$
9. Yield strength: $57 \pm 3 \text{ MPa}$
10. Elongation at break: $\geq 50\%$
11. Impact strength at 23°C: $\geq 5.5 \text{ kJ/m}^2$; at -40°C: $\geq 4.0 \text{ kJ/m}^2$
12. Temperature working range: -40°C to 85°C
13. Shrinkage after 120°C heating: $< 0.5\%$

Are all of these requirements correct for your application? Please confirm or let me know if any need adjustment.

User: looks good recommedn the grades

Assistant: Based on your requirements, here are the top grade suggestions for the loose tube inside your optical fiber cable for indoor data centers:

1. CELANEX® 2001 (Natural) – Hydrolysis-resistant, purpose-built for optical fiber buffer tubes, excellent extrusion performance, very low water uptake, and robust mechanical

properties.

2. Crastin® S600F10 NC010 (Natural) – Proven loose tube grade with strong historical use, balanced toughness and flexibility, and meets all critical CTQs.
3. CELANEX® 1700A (Natural) – High melt strength PBT for extrusion, very low water absorption, and robust mechanical properties.
4. CELANEX® 1600A (Natural) – Cost-effective, high MW PBT extrusion grade, suitable for mild indoor environments.

These grades may be suitable options for your application, but final material selection and product design decisions should be based on your specific requirements and testing. Please ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.